Comparing Agent-based and DSGE Models –An Indirect Inference Estimation Approach–

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Abstract

We propose a new approach for estimating macroeconomic agent-based (AB) models that keeps the mainstream family of dynamic stochastic general equilibrium (DSGE) models as benchmark. We consider a New Keynesian DSGE model (Galí, 2008), assume autonomous interacting agents with bounded rationality and derive an AB model that is simulated to obtain artificial data. The parameters of the AB model are estimated with the method of indirect inference (Gourieroux and Monfort, 1997), where, instead of a simple VAR, the auxiliary model is the state space form representation of the DSGE model. This allows to map the structural parameters of the AB model into the structural parameters of the DSGE model, and to measure the degree of similarity or discrepancy between the two models. Hence, we can tackle issues such the following: Which DSGE parameters are more affected by relaxing the hypothesis of perfect rationality? Do AB models fit real data better than DSGE models of comparable complexity?

Keyword: Agent-based models, DSGE models, Indirect inference

References

[1] Galí, J., "Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework," Princeton University Press, 2008.

[2] Gourieroux, C. and A. Monfort, "Simulation-based Econometric Methods," Oxford University Press, 1997.